

Press Release (PR)

Mercedes-Benz Introduces MBQX, an AI-Powered Intelligence Platform for Production Quality Improvement

Mercedes' new platform unifies fragmented quality systems into a single intelligent workspace, with a generative AI assistant that cuts fault-case resolution time from days to hours, constantly improving the production process

STUTT GART, Germany, September 24, 2026, Mercedes-Benz Group AG (FWB: MBG) today announced the general availability of MBQX, an AI-powered quality intelligence platform that connects every relevant Mercedes-Benz quality system into one workspace and helps production quality engineers resolve fault cases faster, more consistently, and with greater confidence. MBQX combines digital fault records, end-to-end views across the product lifecycle, and a generative AI assistant, the QX Assistant, that actively guides engineers through investigation and root-cause analysis. In a six-month trial, MBQX was found to reduce the number of faults in production by 36%, and decrease the time to fault resolution by 95%.

Today, when a quality issue is reported from the field, for example, a noisy or ineffective windshield wiper, a Mercedes-Benz production quality engineer must navigate up to nine disconnected systems, each with separate credentials and access reviews, manually gather data from more experienced colleagues across departments, and consolidate findings from incomplete data into static reports under a tight deadline. Across Mercedes-Benz plants worldwide, quality engineers manage approximately XX fault cases per month, spending an estimated XX% of their investigation time on data collection and single issue resolution rather than root-cause analysis on issues that all have the same underlying cause. The result is duplicated work, decisions made without complete information, and delayed resolution, costing the business €XXm a year in lost production volume.

MBQX is designed to change that. It is not a replacement for existing systems but an intelligent networking layer that bidirectionally connects them, turning fragmented, locally stored quality data into a single, searchable, and actionable knowledge base. MBQX gives quality engineers a unified workspace where a pre-analyzed investigation brief is ready the moment a case is opened, the institutional knowledge of every prior case is searchable in seconds, and end-to-end visibility across the product lifecycle ensures no information is lost between departments. The result is a quality production process that resolves faults in hours rather than days, builds knowledge rather than losing it to personnel changes, and makes decisions on complete data.

In early deployments across XX Mercedes-Benz plants, engineers report investigation cycles compressing from a two-day deadline to a matter of hours, with the QX Assistant pre-analyzing supplier claims, body measurements, part-version history, and comparable historical cases before the engineer opens the case. "Before MBQX, I would spend my entire two-day deadline just collecting data from different systems and waiting for colleagues to get back to me," said Robin, Production Quality Engineer at Mercedes-Benz Plant Bremen. "Now I get a pre-analyzed investigation brief the moment I open a case. I spend my time understanding the problem and fixing it, not searching for information."

MBQX delivers four core capabilities to quality teams across Research and Development, Mercedes-Benz Operations, and Mercedes-Benz Sales. At its foundation is a unified frontend, a single, intuitive web application accessed through Mercedes-Benz Single Sign-On, that connects all relevant quality systems, including Q LIVE, MRS, AQUA, ZEUS, 8D, Malfunction, Q-Dome, Report X, and IQVis, removing the need for separate access requests across each. Built on top of this unified access layer are digital fault records called QX-Cases, persistent, searchable case files that consolidate every piece of information related to a fault, replacing manual document handoffs and building an institutional knowledge base that survives personnel changes. The QX Assistant takes this further by recognizing the failure scenario, automatically executing digital checks across connected systems, identifying patterns from historical cases, suggesting next steps, guiding the engineer through remaining physical verifications, and drafting the final investigation report. End-to-end views complete the platform, giving teams comprehensive visibility into faults, parts, and vehicles across the full product lifecycle, from development through production to the field, so that no information is lost between departments. MBQX is deployed in the AWS Europe (Frankfurt) Region and aligned with the AWS Well-Architected Framework, supporting Mercedes-Benz's stringent requirements for security, data residency, and regulatory compliance. The QX Assistant uses leading foundation models accessed through Amazon Bedrock, with pattern-detection models trained on decades of historical investigation data and cross-system retrieval powered by managed AWS services.

"Our quality data was like islands, valuable information stored locally, untransparent, and inaccessible across departments," said Jan Terfehr, Product Owner MBQX, Mercedes-Benz Group AG. "MBQX connects what belongs

50 together. With the QX Assistant, we go further: we do not just make information findable, we make it actionable. The
51 platform handles the data work so our engineers can focus on the thinking, interpreting findings, performing physical
52 checks, and getting the right fix into the field faster."

53 MBQX is generally available today to quality teams across Mercedes-Benz, with planned roadmap milestones including
54 expanded failure-scenario coverage for the QX Assistant, deeper integration with process-mining tools for predictive
55 analytics, and progressive rollout to additional plants worldwide.

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58 FAQs

59 Customer FAQs

60 FAQ 1: What is the customer problem?

61 Mercedes-Benz production quality engineers must resolve fault cases, defects or failures reported from the
62 field, within a fixed two-day deadline. The core problem is that the data required to investigate any fault is
63 fragmented across up to nine disconnected systems, each requiring separate credentials, access approvals, and
64 manual data exports. Key pain points include:

- 65 • Engineers spend an estimated TK% of their investigation time gathering data rather than analyzing it.
- 66 • Across TK plants, quality teams handle approximately TK fault cases per month under this manual,
67 fragmented process.
- 68 • Decisions are routinely made without complete information, because relevant data sits in systems the
69 engineer cannot access in time.
- 70 • The downstream impact includes higher Warranty and Goodwill (W&G) costs, elevated Post-
71 Production Rework Costs (PNAK), and fixes that do not hold beyond the defined clean point, requiring
72 repeat investigation of the same fault.

73 FAQ 2: Who are the customers?

74 MBQX serves the full quality community at Mercedes-Benz across Research & Development, Mercedes-Benz
75 Operations, and Mercedes-Benz Sales. Users include:

- 76 • **Primary users:** Production Quality Engineers conducting 8D investigations — the role that bears the
77 greatest burden of manual data gathering today.
- 78 • **Operational users:** Pls/BIs, PAIs, Forepersons, BTVs, BRVs, and PF-Teams who contribute to or monitor
79 fault resolution.
- 80 • **Management and oversight users:** Q-Control Centers and Management, who benefit from aggregated
81 insights, end-to-end fault visibility, and improved reporting quality.
- 82 • **Extended users:** Dealers, who interact with quality outcomes in the field.

83 FAQ 3: What is the solution's vision?

84 MBQX is the connecting layer that has been missing, not a replacement for existing systems, but an intelligent
85 networking layer on top of them. The vision is a quality intelligence platform where:

- 86 • Every engineer opens a single workspace to access all relevant quality systems, regardless of plant or
87 department.
- 88 • A pre-analyzed investigation brief is ready the moment a case is assigned.
- 89 • Root-cause analysis is completed in hours, not days.
- 90 • Every resolved case builds an institutional knowledge base, so the organization learns as a whole
91 rather than relying on individual expertise that leaves when engineers change roles.

92 FAQ 4: What is MBQX?

MBQX is an AI-powered quality intelligence platform that connects every relevant Mercedes-Benz quality system into one unified workspace. It delivers four core capabilities:

- **Unified frontend:** A single web application accessed via Mercedes-Benz Single Sign-On, connecting Q LIVE, MRS, AQUA, ZEUS, 8D, Malfunction, Q-Dome, Report X, and IQVis — no separate access requests required.
- **QX-Cases (digital fault records):** Persistent, searchable case files that consolidate all fault-related information in one place, replacing manual document handoffs and building an institutional knowledge base.
- **QX Assistant:** A generative AI assistant that recognizes failure scenarios, executes digital checks autonomously, identifies patterns from historical cases, guides the engineer through physical verifications, and drafts the final investigation report.
- **End-to-end views:** Comprehensive visibility across the full product lifecycle — from Research & Development through production to the field — so no information is lost between departments.

FAQ 5: What is the QX Assistant?

The QX Assistant is a generative AI assistant within MBQX that supports fault case resolution from the moment a case is opened. Here is how it works:

1. The engineer describes the issue in natural language.
2. The QX Assistant recognizes the failure scenario and automatically runs all digital checks it can execute — vehicle history, supplier claims, body specifications, part versions, and historical patterns.
3. Results are presented in a structured investigation flow, with each finding linked to its source system so it can be verified.
4. The QX Assistant identifies which physical checks the engineer must perform.
5. Once the engineer confirms those checks, the QX Assistant generates a complete investigation report.

The engineer is always the decision-maker. The QX Assistant presents findings and recommendations; the engineer validates and approves. The system learns from corrections over time.

[Mercedes-Benz FAQs](#)

FAQ 6: How will success be measured?

Four primary metrics govern MBQX's success:

- **Cleanpoint-Einhaltung rate:** The share of fixes that hold from the defined clean point. Target: TK% improvement over the baseline of TK%.
- **Time for Problem Resolution:** End-to-end cycle time from fault assignment to closed case. Target: reduction from TK hours to TK hours.
- **IR-Rate:** A composite score covering resolution time, cleanpoint adherence, and recommendation quality.

- **Gamification score:** Individual and team performance improvement tracked over time.

Secondary metrics include reductions in Warranty and Goodwill (W&G) costs and Post-Production Rework Costs (PNAK).

FAQ 7: What is the North Star vision?

The North Star for MBQX is a Mercedes-Benz in which no quality engineer ever opens the same problem twice. The long-term trajectory is:

- **Near-term:** Every engineer resolves fault cases in hours rather than days, with a pre-analyzed brief ready at case open.
- **Medium-term:** The QX Assistant detects emerging failure patterns before they become systemic, shifting the quality organization from reactive investigation to proactive prevention.
- **Long-term:** Within TK years, MBQX supports TK engineers across TK plants globally, reducing enterprise-wide W&G costs by TK and PNAK by TK.

FAQ 8: What data informs this vision?

The case for MBQX is grounded in direct observation of engineer workflows and system audits. Key data points include:

- Engineers consistently reported spending the majority of their two-day investigation window on data collection rather than analysis — corroborated by TK [user study / time-motion analysis — TK reference].
- Historical fault case data spanning TK years of 8D investigation records confirms that TK% of cases involve data from three or more systems.
- TK% of re-opened cases cite incomplete initial data as a contributing factor.
- Customer satisfaction and warranty cost trends further validate the urgency: TK [insert relevant trend data].

FAQ 9: What are the biggest risks, and how will they be mitigated?

Three primary risks have been identified:

- **Data quality:** The QX Assistant's pattern recognition depends on the quality of historical investigation data. Mitigation: a dedicated data-quality review before model training, and ongoing human feedback loops to correct errors.
- **Adoption:** Engineers accustomed to existing workflows may be slow to change. Mitigation: Key User champions in each plant, introductory webinars, and a phased rollout that allows teams to build confidence before broader deployment.
- **Integration stability:** Bidirectional connections to nine source systems introduce dependency risk. Mitigation: API health monitoring, graceful degradation so MBQX remains usable if a source system is temporarily unavailable, and a defined escalation path for integration failures.

FAQ 10: How could MBQX go bigger?

163 MBQX is currently scoped to fault rectification within the quality organization. Three expansion paths are
164 possible:

- 165 • **Supplier quality management:** Extend the intelligence layer to enable proactive supplier performance
166 monitoring and earlier intervention before defects reach production.
- 167 • **Over-the-air fix integration:** Connect MBQX to vehicle software update pipelines, allowing field quality
168 signals to trigger over-the-air fixes without physical plant intervention for software-related faults.
- 169 • **Full lifecycle operating system:** At full scale, MBQX becomes the quality operating system for
170 Mercedes-Benz across the entire vehicle lifecycle — from development through to the field.

171 **FAQ 11: What signals are absent that may be meaningful?**

172 Three absent signals are worth monitoring:

- 173 • **End-customer voice:** MBQX has no direct feedback loop from vehicle owners. Warranty claims and
174 field reports enter through dealer and MS systems, but unstructured customer feedback is not yet
175 integrated, a gap that may cause the platform to optimize for internal process speed while missing
176 emerging quality perceptions in the market.
- 177 • **Supplier perspective:** Supplier root-cause data is available for analysis, but suppliers do not yet
178 interact with MBQX directly. Their view on why defects reach Mercedes-Benz is currently filtered
179 through internal records.
- 180 • **Non-adopters:** There is no signal yet from engineers who choose not to use the QX Assistant.
181 Understanding resistance patterns early will be important for adoption planning.

182 **FAQ 12: What are the hotly debated topics?**

183 **How should MBQX handle cases where the QX Assistant's recommendation conflicts with an engineer's prior**
184 **experience?**

185 The dilemma is that experienced engineers have tacit knowledge that the model may not capture, but over
186 time, allowing engineers to routinely override AI recommendations without structured feedback will degrade
187 the model's ability to learn from disagreement. The current recommendation is to require engineers to log a
188 reason when they override a QX Assistant recommendation, creating a structured feedback dataset. The open
189 question is whether this requirement will be perceived as punitive or bureaucratic, and how to design the
190 logging experience so it feels like contribution rather than compliance. Input from the design and change
191 management teams is needed before the override workflow is finalized.

192 **FAQ 13: Who will MBQX collaborate with to achieve this?**

193 MBQX requires sustained collaboration across multiple organizations. Key dependencies include:

- 194 • **IT organization:** SSO integration and source system API access.
- 195 • **Legal and Data Privacy:** Ongoing compliance review as new data sources are connected.
- 196 • **Quality organization leadership:** Adoption sponsorship and Key User nomination in each plant.
- 197 • **Analytics team:** Celonis integration for advanced process mining.
- 198 • **AWS:** Cloud infrastructure and managed AI services underpinning the QX Assistant.

- **Source system owners:** Q LIVE, MRS, AQUA, ZEUS, 8D, Malfunction, Q-Dome, Report X, and IQVis system owners are standing stakeholders whose cooperation is required for any integration change.

FAQ 14: What other options were considered?

Three alternatives were evaluated before selecting MBQX's architecture:

- **Expanding Celonis** as the primary quality intelligence layer was rejected because core fault rectification workflows should not depend on a single third-party commercial platform, and Celonis's strength is process mining rather than case management.
- **Point-to-point integration** between the two highest-burden systems (AQUA and MRS) was rejected because it addresses two of nine integration gaps and does not solve the fragmentation problem structurally.
- **A lightweight data warehouse with a reporting frontend** was rejected because static reporting does not support the real-time, case-level investigation workflow that engineers need.

MBQX's architecture — a bidirectional intelligent networking layer with a generative AI assistant, was selected because it addresses both the access problem and the analysis problem simultaneously.

FAQ 15: How does MBQX accelerate Mercedes-Benz's quality and customer satisfaction strategy?

MBQX directly accelerates Mercedes-Benz's commitment to building vehicles that meet the expectations of premium and luxury customers. Key contributions include:

- **Faster fault resolution:** Reduces the time between a quality issue entering the field and a durable fix reaching production, lowering warranty exposure and improving the ownership experience.
- **Compounding quality improvements:** The institutional knowledge base built through QX-Cases means quality improvements accumulate over time rather than resetting with each personnel change, a structural advantage for a company that competes on engineering reputation.
- **Sustainability impact:** Reduces rework, scrap, and repeat manufacturing effort associated with unresolved or misdiagnosed faults, supporting Mercedes-Benz's broader sustainability obligations.

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Visuals

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(image pending)

[Customer experience begins here, with the problem / challenge]

2

(image pending)

[Customer longs for a better way and discovers solution]

3

(image pending)

[Customer explores solution and signs up]

4

(image pending)

[Customer begins to onboard / engage with new solution]

5

(image pending)

[Customer discovers new feature that makes their life easier]

6

(image pending)

[Customer discovers new feature that makes their life easier]

7

(image pending)

[Customer discovers new feature that makes their life easier]

8

(image pending)

[Customer is left happy and satisfied]